The Key Challenges and Technological Innovations in BGP

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2025.09

Content

- **Development History of BGP**
- Key Challenges in BGP
- Technical Innovations in BGP
- Toward the Next-Generation BGP

The Development History of BGP

Phase I

BGP-1
RFC 1105

- Based on the path vector protocol
- Records the AS sequence in the path
- Supports multi-homed networks

- Initial version

1989

Phase II

BGP-2
RFC 1163

- Enhanced message types (such as Keepalive) to improve stability

- Insufficient routing scalability
- Difficult to meet the demands of larger-scale networks

1990

Phase III

BGP-3
RFC 1267

- Support for more complex AS topologies
- improving routing update efficiency

- Scalability is still insufficient
- Routing policy supports only simple configurations

1991

Phase IV

BGP-4
RFC 1771

- Support for CIDR and route aggregation
- Support for Multi-Protocol Extensions (MP-BGP)
- Significantly enhances routing policy flexibility
- In 2006, RFC 4271 replaced RFC 1771
- Since 2006, BGP has not undergone any major version updates

- There is still significant room for improvement in terms of security, performance, and other aspects

1995

Significant improvement in scalability, flexibility, and stability

Content

- Development History of BGP
- **Key Challenges in BGP**
- Technical Innovations in BGP
- Toward the Next-Generation BGP

Key Challenges in BGP

Security

- Routing security risks **rooted in the original designs** of inter-domain routing protocols.
- The **lack of a built-in trust model** has led to persistent routing security vulnerabilities and hindered effective coordination and optimization of routing policies.

Manageability

- BGP **misconfigurations** by network operators frequently result in service disruptions.
- Inter-domain routing systems often suffer from **delayed anomaly detection** and **high recovery latency**, leading to prolonged network instability.

Scalability

- With the continuous expansion of the Internet, **the rapid growth of routing table size** is placing growing pressure on the performance of inter-domain routing systems.
- The **rigid and static semantics of BGP limit** its ability to adapt to the evolving demands of emerging services.

Security Challenges

Global Routing Security Incidents

- **2008.02.23** — Pakistan Telecom hijacked YouTube prefixes, causing global service disruption
- **2017.04.26** — Russia's Rostelecom hijacked prefixes of major international financial websites
- **2018.04.24** — 1300 IPs belonging to Amazon were hijacked for 2 hours to steal cryptocurrency
- **2019.06.24** — Verizon route leak caused widespread outages affecting AWS and other cloud services
- **2020.04.01** — Rostelecom suspected of hijacking Google and AWS traffic
- **2022.09.25** — Hackers hijacked 256 Amazon IPs via BGP, stealing $1.68 million in crypto assets
- **2024.01.05** — Spanish ISP Orange suffered a 3-hour outage due to hijacked ASN and invalid RPKI configuration
- **2024.06.27** — Cloudflare experienced BGP hijack and route leak, impacting DNS service (1.1.1.1) across 300 networks in 70 countries

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Network Working Group                                          S. Murphy
Request for Comments: 4272                                 Sparta, Inc.
Category: Informational                                     January 2006

                  BGP Security Vulnerabilities Analysis

Status of This Memo

   This memo provides information for the Internet community.  It does
   not specify an Internet standard of any kind.  Distribution of this
   memo is unlimited.

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Abstract

   Border Gateway Protocol 4 (BGP-4), along with a host of other
   infrastructure protocols designed before the Internet environment
   became perilous, was originally designed with little consideration
   for protection of the information it carries.  There are no
   mechanisms internal to BGP that protect against attacks that modify,
   delete, forge, or replay data, any of which has the potential to
   disrupt overall network routing behavior.

   This document discusses some of the security issues with BGP routing
   data dissemination.  This document does not discuss security issues
   with forwarding of packets.
```

RFC4272 BGP Security Vulnerabilities Analysis

This **inherent vulnerability of BGP** becomes particularly pronounced in the face of malicious attacks or misconfigurations, posing significant threats to nations, network operators, and enterprises.

VI.1.3.1 Threats to Internet Routing

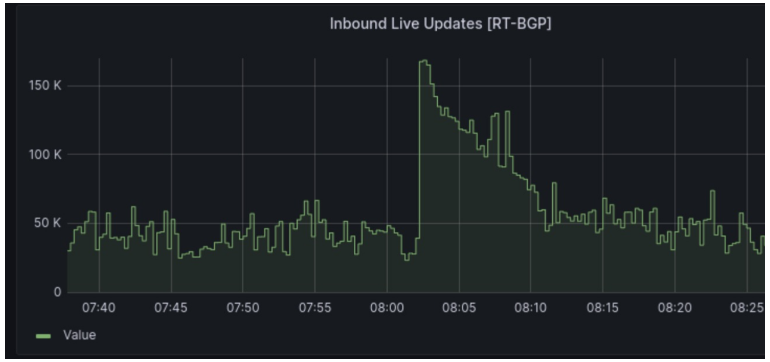
What threats to Internet routing should the Commission consider within the scope of this inquiry in addition to BGP hijacking? For example, to what extent could BGP security measures prevent pervasive monitoring?

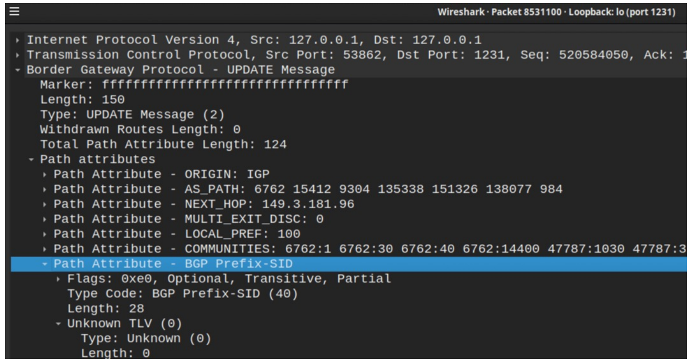
We consider BGP hijacking, route leaks, and address spoofing in scope. For clarity we provide

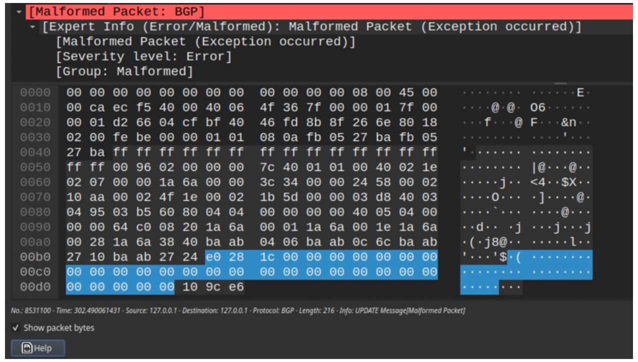
ISOC identifies **three core threats** to Internet routing security: **BGP hijacking**, **route leaks**, and **address spoofing**.

Manageability Challenges

- ✓ **The butterfly effect of (small) misconfigurations**: Such misconfigurations can trigger severe security incidents. In extreme cases, they may result in global-scale Internet disruptions.
- ✓ **Routing anomaly detection and recovery remain challenging**: Fragmented global risk visibility and limited real-time monitoring hinder timely detection. Lack of policy synchronization across Autonomous Systems further increases recovery complexity and delay.





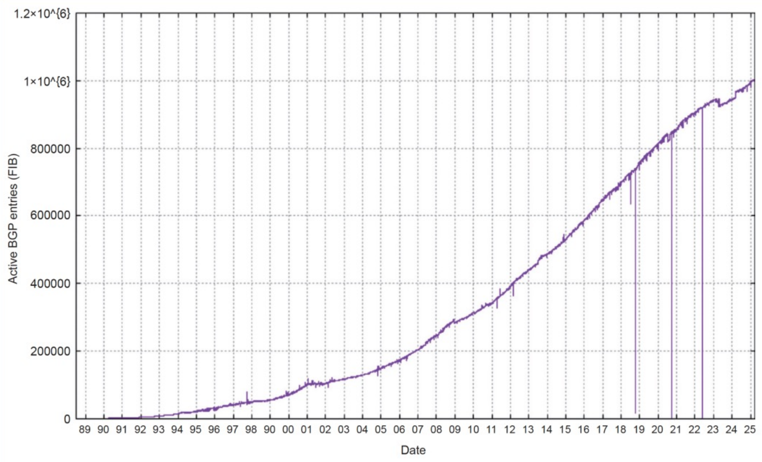


On May 20, 2025, a global Internet routing instability event occurred **due to a BGP misconfiguration. Approximately 100 seperate networks were affected**, including well-known networks such as SpaceX Starlink (AS14593), ByteDance (AS396986), and Disney Global Services (AS23344).

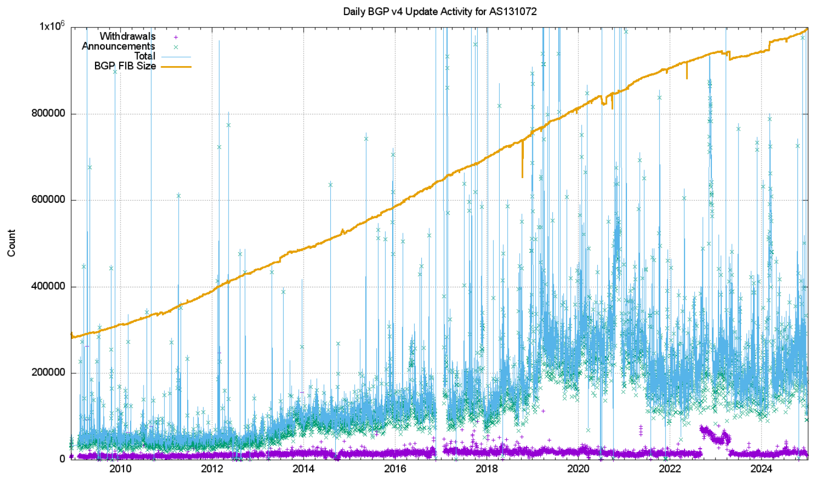
Lack Automated Configuration, Intelligent Anomaly Detection, and Self-recovery Mechanism

Scalability Challenges

- ✓ **Routing table growth leads to performance bottlenecks**: The continuous expansion of routing tables, along with high-frequency BGP updates, exacerbates routing instability. This not only significantly prolongs convergence time but also increases network hardware costs.
- ✓ **Growing structural incompatibility**: BGP’s legacy design is misaligned with modern network requirements. The protocol’s rigidity increasingly clashes with the dynamic evolution of emerging technologies.



Rapid growth of routing table entries

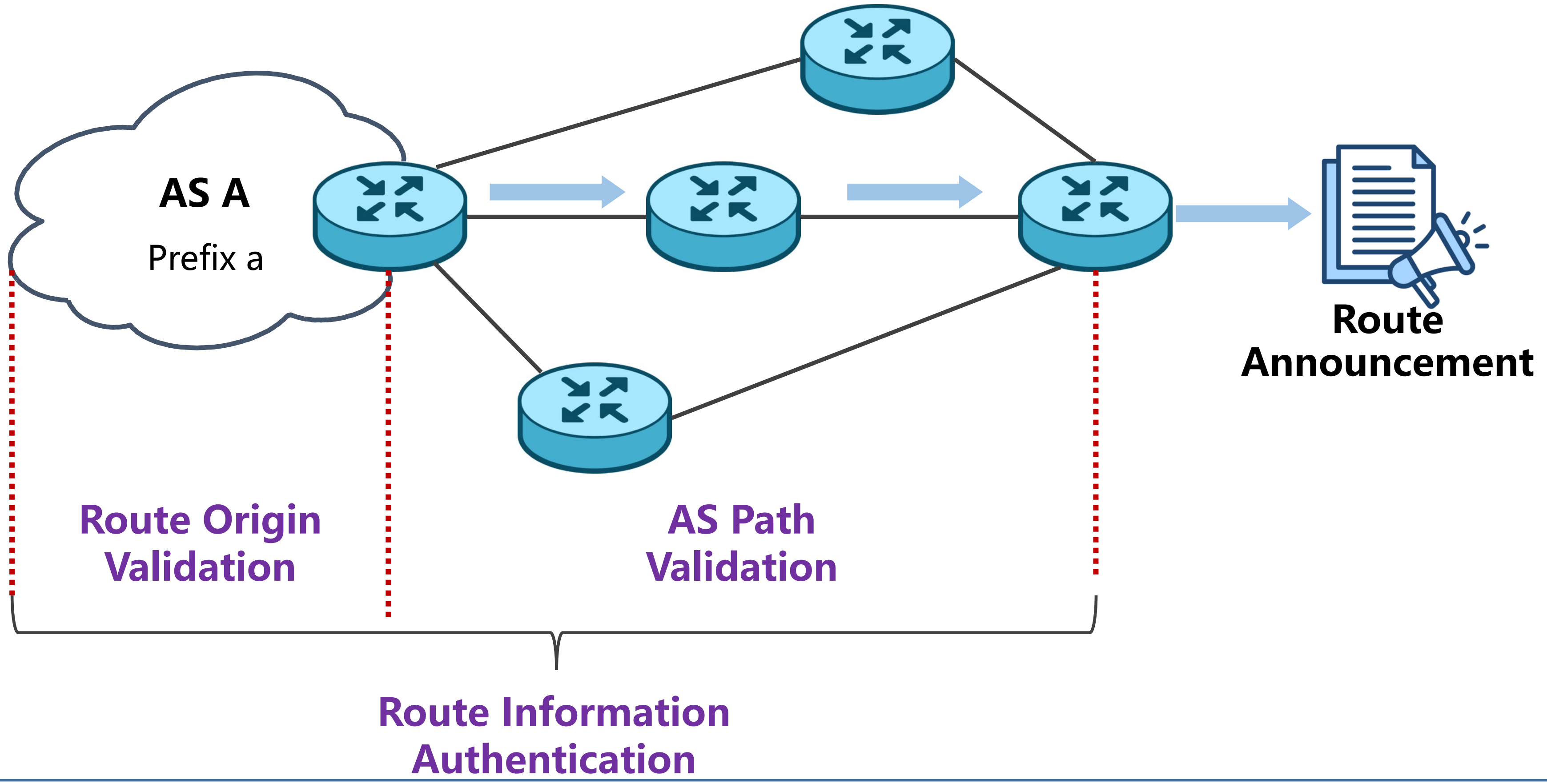


High-frequency routing updates

Insufficient capability to advance protocol evolution without compromising global routing stability

Content

Security



- ✓ **Route origin validation** and **announcement path validation** form the foundation of the current **routing security framework**.
- ✓ These two mechanisms are **represented by RPKI and BGPsec**, respectively.

RPKI

2006

- **SIDR WG established**
- **The standardization of the RPKI started**

2012-2014

SIDR WG

RPKI: RFC 6480 ~6493, 6810, 6811, 6907, 6916, 6945, 7115, 7128, 7318

2016

SIDROPS WG established

- Deployment and operation of inter-domain routing security

2016-2018

- **SIDR WG**
- RPKI: RFC7382, 7730, 7909, 7935, 8097, 8181, 8182, 8183, 8210, 8211, 8360, 8416

2018-2024

- **SIDROPS WG**
- RPKI: RFC8481, 8488, 8630, 8893, 8897, 9255, 9286, 9319, 9323, 9324, 9589, 9674, 9691, 9697

✓ RPKI leverages a public key infrastructure to cryptographically verify the association between IP prefixes and ASNs. This **enables routers to validate the authenticity of BGP announcements through ROV**.

✓ However, the reliance on **centralized trust anchors** introduces a **potential vulnerability in the trust model**.

Routing Information (RPKI)

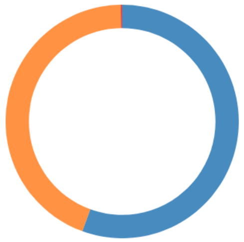
Valid	718,324	56.4%
Unknown	547,018	43.0%
Invalid	8,168	0.6%

Valid Unknown Invalid

Routing Information (RPKI) States

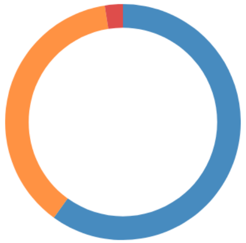
IPv4

Valid	577,394
Unknown	459,779
Invalid	2,282
Total	1,039,455



IPv6

Valid	141,016
Unknown	88,315
Invalid	5,815
Total	235,146



Valid Unknown Invalid

Route Origin Authorizations (ROAs) Stats

Source: MANRS Observatory by June 30, 2025

BGPsec

2006
- **SIDR WG Established**
- Focus on route origin validation and path validation

2014-2016
- **SIDR WG**
- **BGPsec**: RFC 7132, 7353

2016
- **SIDROPS WG**
- Guide the deployment and operation

2016-2018
- **SIDR WG**
- BGPsec: RFC 8205~ 8209

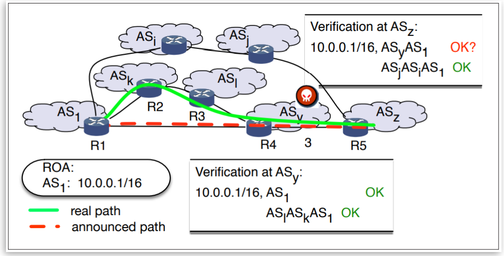
2018-2024
- **SIDROPS WG**
- BGPsec: RFC 8608, 8634, 8635

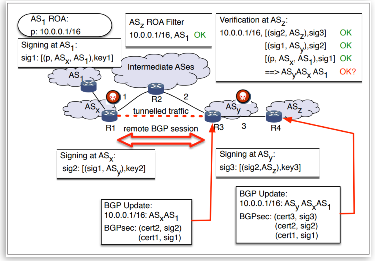
✓ BGPsec aims to address AS path manipulation in inter-domain routing by providing cryptographic validation.
✓ BGPsec **does not support partial deployment**, resulting in **low adoption willingness among ISPs**.


acmqueue
Why Is It Taking So Long to Secure Internet Routing?
Routing security incidents can still slip past deployed security defenses
Sharon Goldberg, Boston University

Sharon Goldberg
Professor
Boston University

Current BGP security mechanisms are **difficult to deploy** and generally **do not support incremental deployment**.





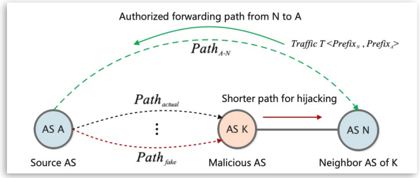
Even with full deployment, BGPsec remains vulnerable to attacks such as **wormhole** and **tampering**.

VRO and FC–BGP

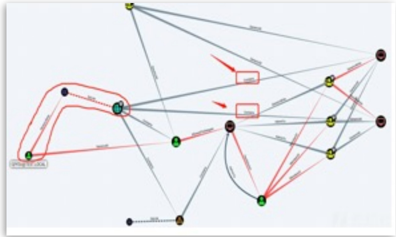
- ✓ **Origin validation**, represented by RPKI, has problems such as data inconsistency, delayed updates, and limited coverage. In addition, the centralized structure has potential trust risks.
- ✓ **Path validation**, represented by BGPsec, requires a high deployment rate. In the scenario of partial deployment of paths, it is difficult to guarantee the security benefits of deployed ASes.

Verifiable Route Origin
VRO

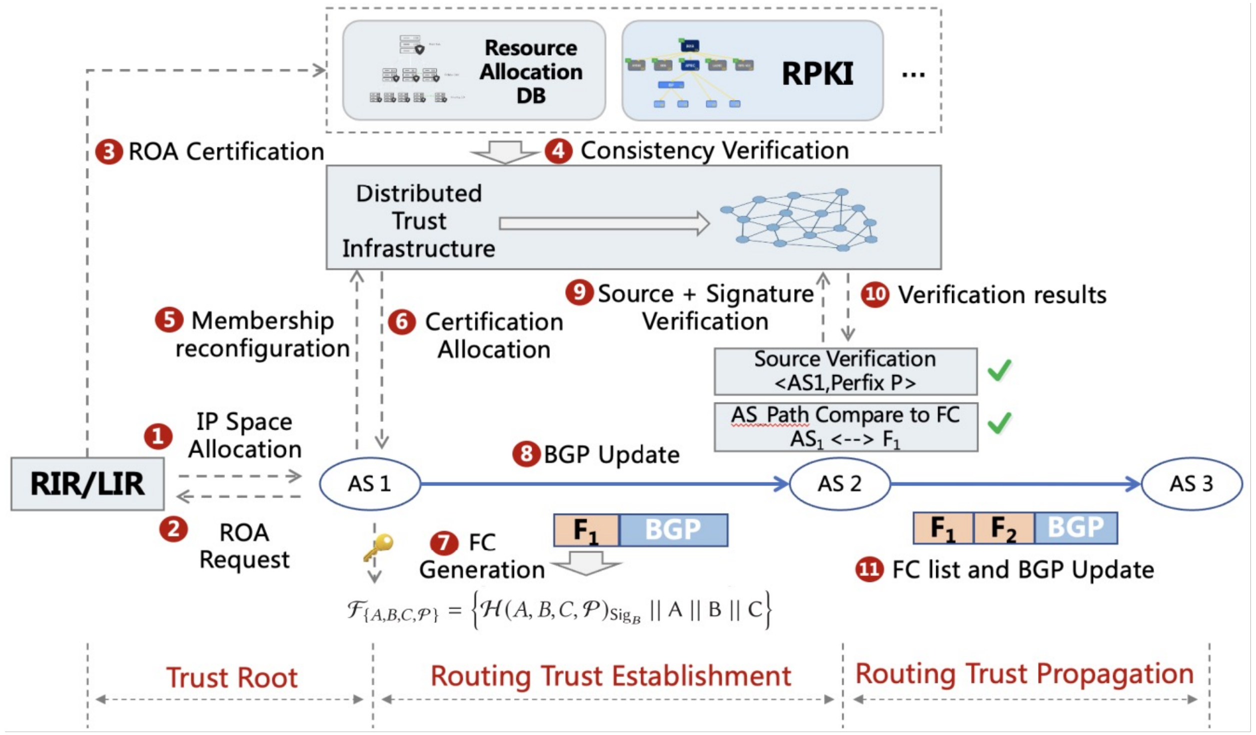
Forwarding Commitment-based BGP
FC-BGP



ROV



Path Validation



- ✓ Through distributed multi-source validation, data source conflicts are resolved, and a source validation trust root is constructed.
- ✓ BGP and forwarding path validation can be achieved through the generation and delivery of FC

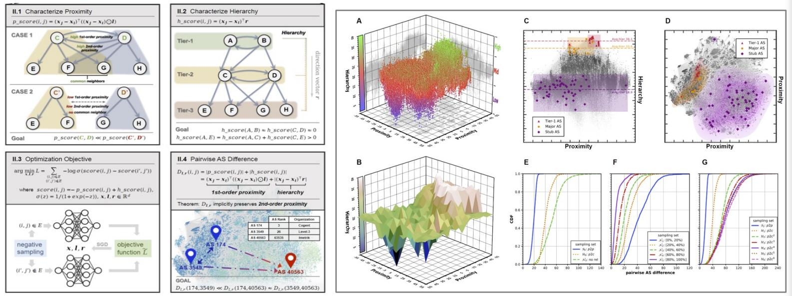
Industry Practices in Routing Anomaly Detection

RIPE NCC
Routing Information Service (RIS)

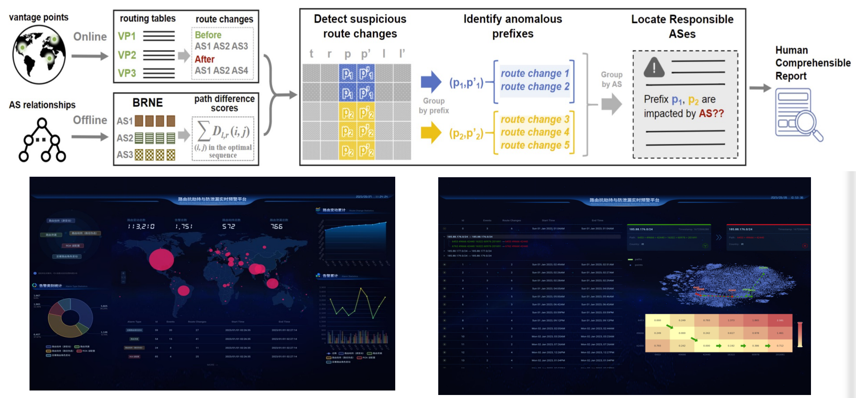
- ✓ Prominent routing data collection platforms include: **RouteViews** and **CAIDA** in the United States, **RIPE RIS** in Europe.
- ✓ Commercial platforms such as: **Cisco ThousandEyes** and **Noction IRP** provide advanced capabilities for routing anomaly monitoring and management.
- ✓ Major Chinese operators:
  China Telecom provides systematic protection and rapid anomaly response.
  China Unicom supports global, real-time cross-domain risk evaluation.
  China Mobile strengthens proactive detection with an integrated routing security knowledge base.

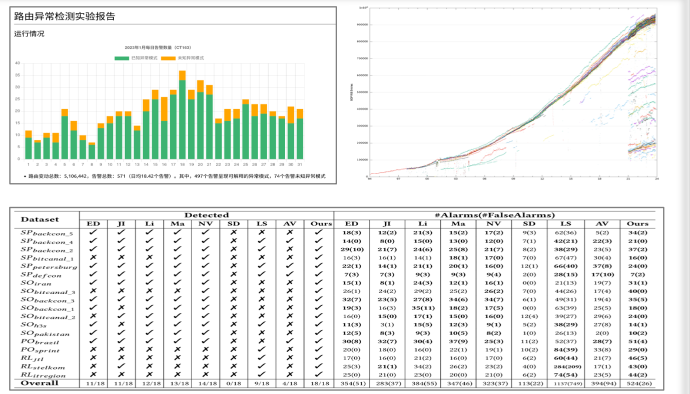
Towards a Semantics–Aware Routing Anomaly Detection System

- ✓ The presence of large-scale, non-deterministic route announcements **makes it challenging to detect routing anomalies in a timely manner**.
- ✓ Identify route hijacks and leaks in real time **by anomaly mechanisms and extracting key BGP attributes**.



Modeling Key BGP Attributes for Routing Analysis and Detection





Inter-Domain Routing Anomaly Feature Representation

By extracting key BGP routing attributes and defining AS routing roles, **various anomalies can be uniformly represented as routing changes** that lead to significant role transitions.

Routing Anomaly Detection and Visualization Monitoring Platform

Developed **a BGP anomaly detection and visualization platform** with real-time output, supporting interpretation of route hijack and leak events to enable comprehensive routing security situational awareness.

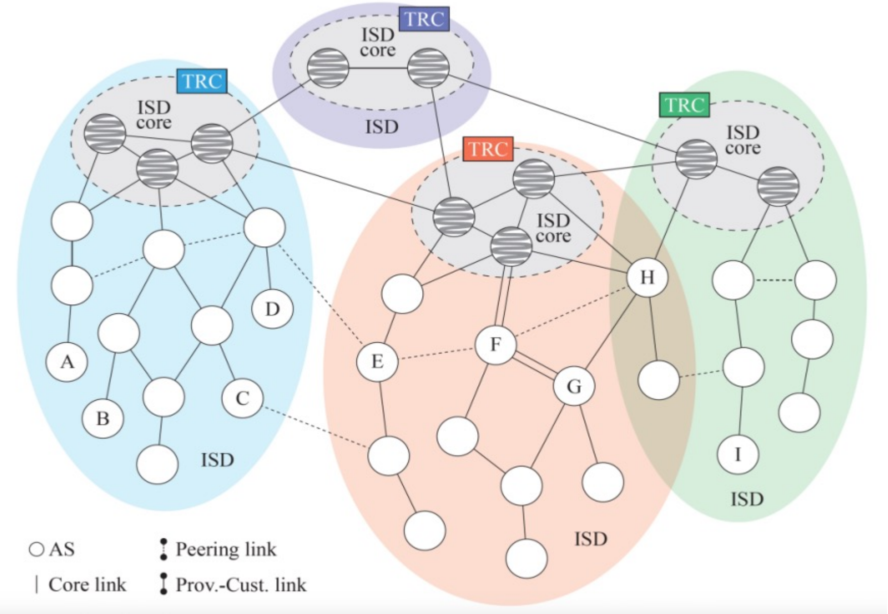
Real-World Routing Deployment and Validation

System deployed in real-world routing environments for continuous anomaly detection and automated monthly reporting.

Learning with Semantics: Towards a Semantics-Aware Routing Anomaly Detection System, USENIX Security 2024, **Internet Defense Prize Winner, Distinguished Paper Award.**

Next–Generation Networks

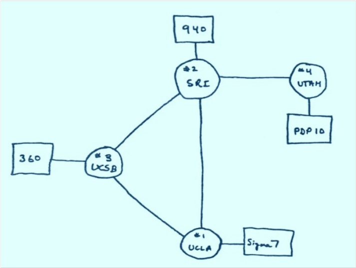
SCION--Scalability, Control, and Isolation on Next-Generation Networks

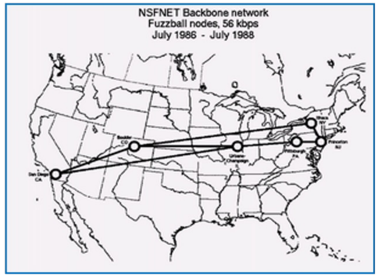


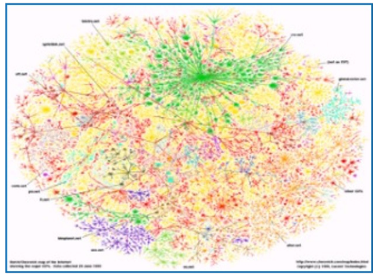
- ✓ SCION proposes a novel network architecture centered around **Isolation Domains(ISDs)**.
- ✓ SCION divides the global Internet into multiple logically isolated domains, each jointly governed by a set of core ASes and **secured via Trust Root Configurations(TRCs)**.
- ✓ Multipath capabilities for **advanced traffic engineering** and trust-based domain architecture for **secure and controllable routing**.

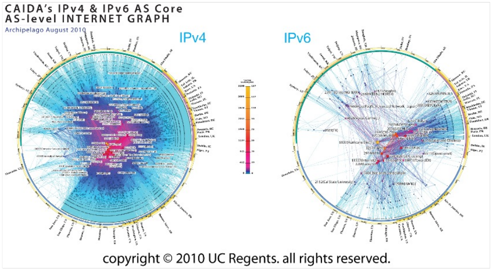
Content

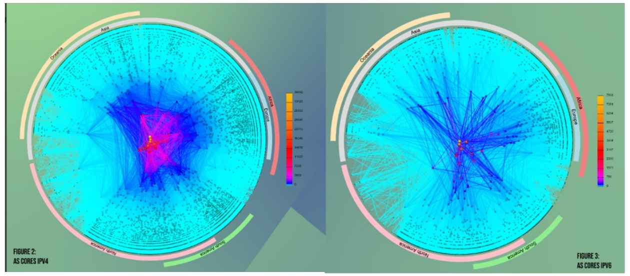
Technical Explorations Behind the Internet’s Success











1969 | 1986 | 1994 | 2010 | 2020

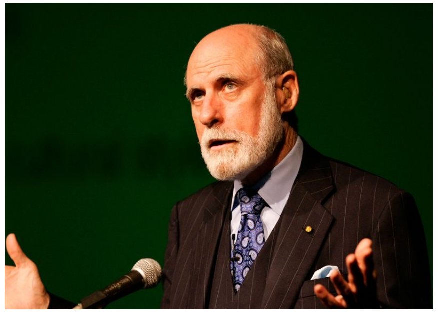
REQUEST FOR PROPOSAL
STUDY ON THE INTERNET'S TECHNICAL SUCCESS FACTORS

From: Asia Pacific Network Information Centre (APNIC), and Registro de Direcciones de Internet para America Latina y Caribe (LACNIC)

Date: 4 March 2021 (Current Version)

Re: **Study on the Internet's Technical Success Factors**

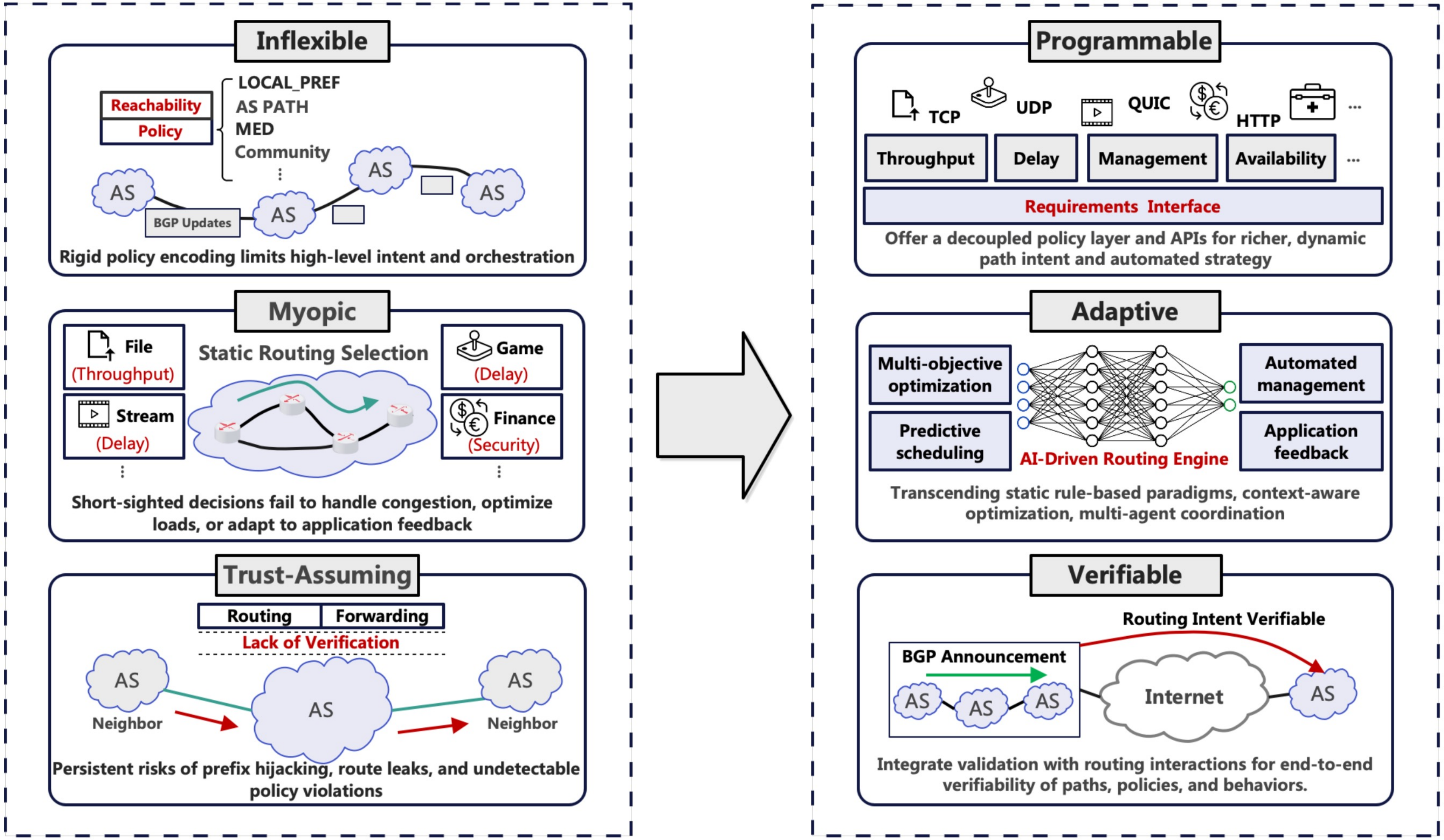
Joint collaborators APNIC and LACNIC invite proposals by individual consultants or aconsortium of consultants to study the Internet‘s Technical Success Factors. The following RFP includes backgrounds of our organisations and describes the desired purpose and specific requests relating to the proposal.



Vint Cerf
Father of the Internet
Openness: The Foundation of Internet Success

Brian E. Carpenter
Former chair of the IETF, IAB, ISOC
Key Enablers: Incremental Design & Backward Compatibility

Toward the Next-Generation BGP



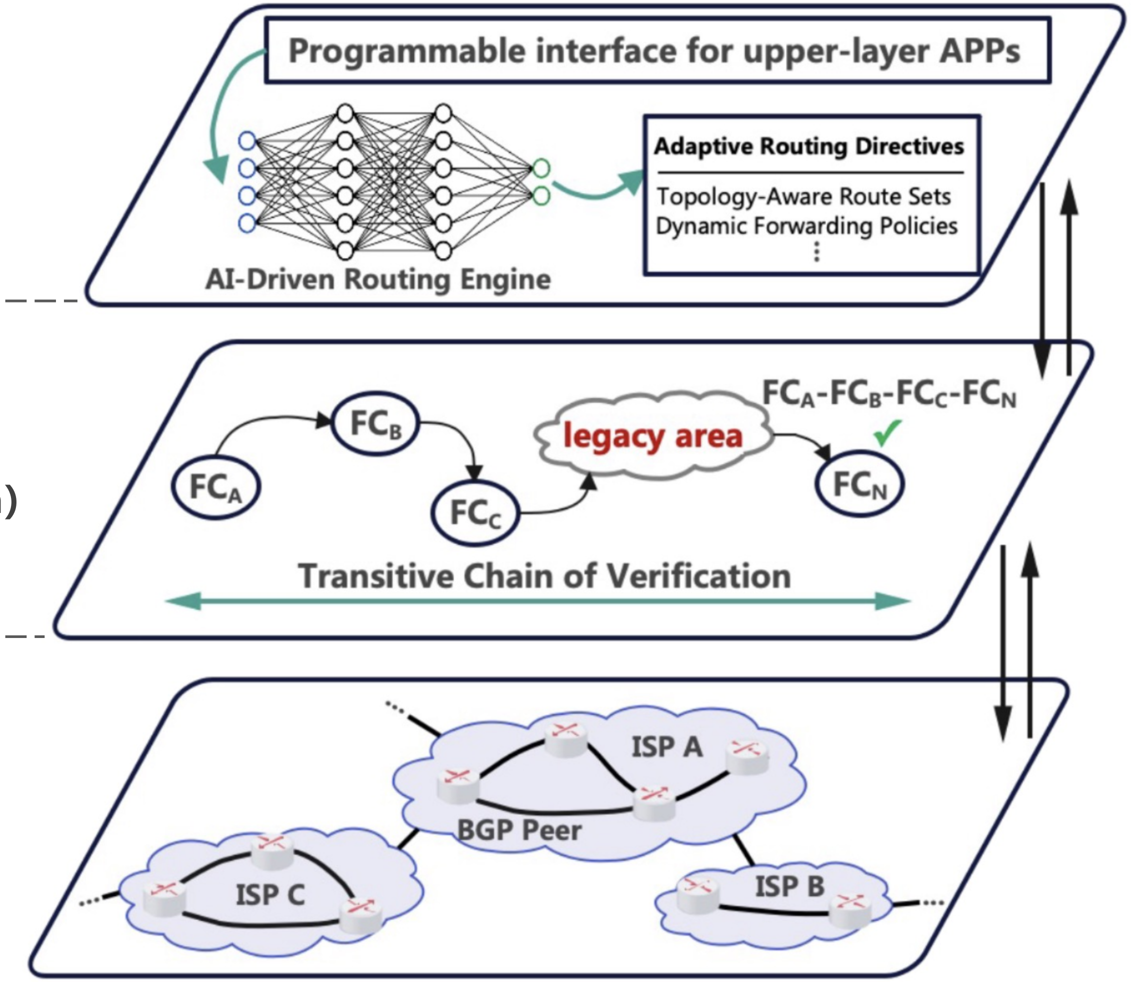
Towards a **Secure**, **Intelligent**, and **Policy-Decoupled** Inter-Domain Routing

Toward the Next-Generation BGP

Policy Plane
(Programmable, AI-Driven Optimization)

Native Trust
(Forwarding Commitments and Verification)

Connectivity Plane
(Stable Reachability Substrate)



The next-generation BGP architecture **follows a layered model**, comprising: **connectivity plane**, **embedded trust plane** and **policy plane**

Toward the Next-Generation BGP

Connectivity Plane

Ensures basic inter-AS reachability by establishing and maintaining minimal viable paths across the Internet. This plane performs no policy-based path selection, and functions analogously to a "clean forwarding substrate" upon which higher-level policy decisions can be enacted.

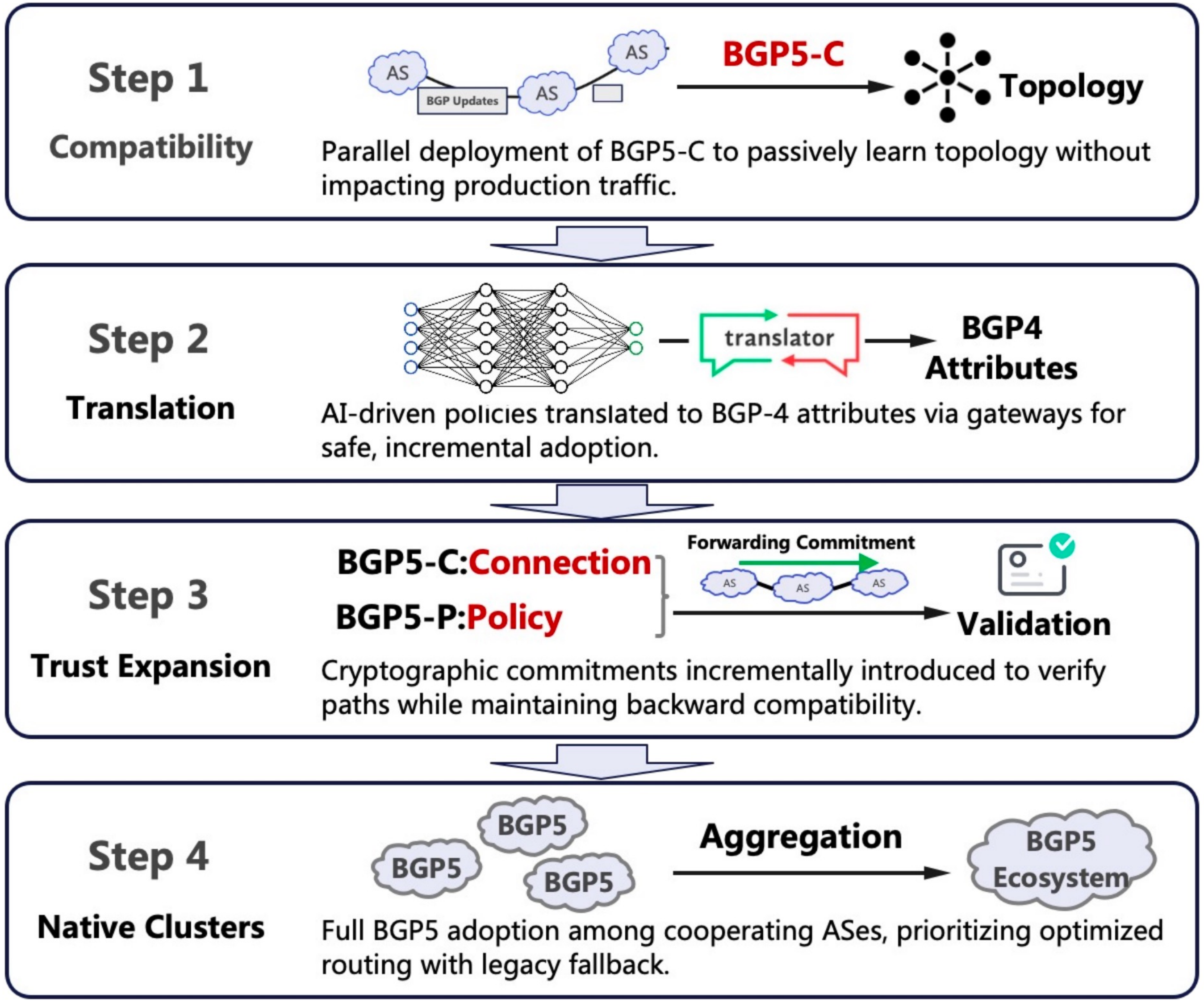
Native Trust Plane

Implements verifiable forwarding commitments (FCs) and path authentication mechanisms. This layer **ensures that routing updates and path decisions are provably legitimate and traceable**, even when originating from autonomous and potentially untrusted domains.

Policy Plane

Responsible for path selection, preference expression, and network-wide optimization. This plane is designed to be programmable and AI-compatible, supporting real-time traffic engineering, application-aware routing, and inter-AS policy negotiation.

Toward the Next-Generation BGP



The BGP5 migration follows four stages:

1. BGP5-C passively learns topology without affecting BGP-4
2. AI-driven preferences converted to BGP-4 attributes via gateways
3. Cryptographic commitments incrementally added to verify paths
4. Full BGP5 adoption with legacy fallback

This phased approach **minimizes risk**, **ensures backward compatibility**, and **allows operators to validate each capability**—connectivity, AI policies, trust, and cross-domain optimization—before full deployment.

Thank You!